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Formatted output

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" $f \in C^k$ $\implies$ $\|f\|_{C^k} \leq C \|f\|_{L^2}$," — anonymous user

§ 1. Introduction

A new I/O-agnostic text formatting library was introduced in C++20 ([\[FORMAT\]](#)). This paper proposes integrating it with standard I/O facilities via a simple and intuitive API achieving the following goals:

- Usability
- Unicode support
- Good performance
- Small binary footprint

§ 2. Revision history

Changes since R0:

- Clarified that adding `wchar_t` overloads will be addressed in a separate paper per the UK C++ panel feedback.

§ 3. Motivating examples

Consider a common task of printing formatted text to `stdout`:

C++20	Proposed
<pre>std::cout << std::format("Hello, {}!", name);</pre>	<pre>std::print("Hello, {}!", name);</pre>

The proposed `std::print` function improves usability, avoids allocating a temporary `std::string` object and calling `operator<<` which performs formatted I/O on text that is already formatted. The number of function calls is reduced to one which, together with `std::vformat`-like type erasure, results in much smaller binary code (see § 7 Binary code).

Existing alternatives in C++20:

Code	Comments
<pre>std::cout << "Hello, " << name << "!";</pre>	Requires even more formatted I/O function calls; message is interleaved with parameters; can result in interleaved output.

C#/.NET	<code>Console.Write</code> [DOTNET-WRITE]
COBOL	<code>DISPLAY</code> statement [N0147]
Fortran	<code>print</code> and <code>write</code> statements [N2162]
Go	<code>Printf</code> [GO-FMT]
Java	<code>PrintStream.format</code> , <code>PrintStream.print</code> , <code>PrintStream.printf</code> [JAVA-PRINT]
JavaScript	<code>console.log</code> [WHATWG-CONSOLE]
Perl	<code>printf</code> [PERL-PRINTF]
PHP	<code>printf</code> [PHP-PRINTF]
Python	<code>print</code> statement or function [PY-FUNC]
R	<code>print</code> [R-PRINT]
Ruby	<code>print</code> and <code>printf</code> [RUBY-PRINT]
Rust	<code>print!</code> [RUST-PRINT]
Swift	<code>print</code> [SWIFT-PRINT]

Variations of `print[f]` appear to be the most popular naming choice for this functionality. It is either provided as a free function (most common) or a member function (less common) together with a global object representing standard output stream. Notable exceptions are COBOL, Fortran, and Python 2 which have dedicated language statements and Rust where `print!` is a function-like macro.

We propose adding a free function called `print` with overloads for writing to the standard output (the default) and an explicitly passed output stream object. The default output stream can be either `stdout` or `std::cout`. We propose using `stdout` because it is considerably faster on at least two major implementations (see [§ 6 Performance](#)) and because `print` won't use any formatted output functionality of `ostream`. Since `stdout` doesn't have an associated locale we propose using the current global locale for locale-specific formatting which is consistent with `format`. With `cout` or another explicitly passed stream, the stream's locale will be used. In all cases the default formatting is locale-independent.

Another option is to make `print` a member function of `basic_ostream`. This would make usage somewhat more awkward:

```
std::cout.print("Hello, {}!", name);
```

A free function can also be overloaded to take `FILE*` to simplify migration (possibly automated) of code from `printf` to the new facility.

There are multiple approaches to appending a trailing newline:

- Don't append a newline automatically: `printf` in C and other languages.
- Append a newline but don't format arguments: `puts` in C (inconsistent with `fputs`).
- Have two formatting functions/macros, one that appends newline and another that doesn't: `print/println` in Java, `print!/println!` in Rust, `Printf/Println` in Go, `Write/WriteLine` in C#/.NET.
- Let the user choose a terminating string defaulting to `"\n"` and do limited formatting: `print` in Python and Swift.

We propose not appending a newline automatically for consistency with `printf` and `iostreams`:

```
std::print("Hello, {}!", name);    // doesn't print a newline
std::print("Hello, {}!\n", name);  // prints a newline
```

Additionally we can provide a function that appends a newline:

```
std::println("Hello, {}!", name);  // prints a newline
```

Although `println` doesn't provide much usability improvement compared to `print` with explicit `'\n'`, it has been a frequently requested feature in the `fmt` library ([\[FMT\]](#)).

§ 5. Unicode

We can prevent mojibake in the Unicode example by detecting if the string literal encoding is UTF-8 and dispatching to a different function that correctly handles Unicode, for example:

```
constexpr bool is_utf8() {
    const unsigned char micro[] = "\u00B5";
    return sizeof(micro) == 3 && micro[0] == 0xC2 && micro[1] == 0xB5;
}

template <typename... Args>
void print(string_view fmt, const Args&... args) {
    if (is_utf8())
        vprint_unicode(fmt, make_format_args(args...));
    else
        vprint_nonunicode(fmt, make_format_args(args...));
}
```

where the exposition-only `vprint_unicode` function formats and prints text in the UTF-8 encoding using the native system API that supports Unicode and `vprint_nonunicode` does the same for other encodings. The latter ensures that interoperability with code using legacy encodings is preserved even though `print` is a new API and it is not strictly necessary.

In Visual C++ `is_utf8` will return `true` if both source and literal encodings are UTF-8, which is enabled by the `/utf-8` compiler flag or other means, and `false` otherwise. It can be implemented in a more elegant way using the encoding detection mechanism proposed by [\[P1885\]](#).

This approach has been implemented in the `fmt` library ([\[FMT\]](#)) and successfully tested on a variety of platforms.

Here's an example output on Windows:

```
C:\projects\fmt>cl print-test.cc /I include /utf-8 /EHsc
Microsoft (R) C/C++ Optimizing Compiler Version 19.24.28316 for x86
Copyright (C) Microsoft Corporation. All rights reserved.

print-test.cc
Microsoft (R) Incremental Linker Version 14.24.28316.0
Copyright (C) Microsoft Corporation. All rights reserved.

/out:print-test.exe
print-test.obj

C:\projects\fmt>print-test
Привет, κόσμος!
```

At the same time interoperability with legacy code is preserved when literal encoding is not UTF-8. In particular, in case of EBCDIC, Shift JIS or a non-Unicode Windows code page, `print` will perform no transcoding and the text will be printed as is.

§ 6. Performance

All the performance benefits of `std::format` ([\[FORMAT\]](#)) automatically carry over to this proposal. In particular, locale-independence by default reduces global state and makes formatting more efficient compared to `stdio` and `iostreams`. There are fewer function calls (see [§ 7 Binary code](#)) and no shared formatting state compared to `iostreams`.

The following benchmark compares the reference implementation of `print` with `printf` and `ostream`. This benchmark formats a simple message and prints it to the output stream redirected to `/dev/null`. It uses the Google Benchmark library [\[GOOGLE-BENCH\]](#) to measure timings:

```

#include <cstdio>
#include <iostream>

#include <benchmark/benchmark.h>
#include <fmt/ostream.h>

void printf(benchmark::State& s) {
    while (s.KeepRunning())
        std::printf("The answer is %d.\n", 42);
}
BENCHMARK(printf);

void ostream(benchmark::State& s) {
    std::ios::sync_with_stdio(false);
    while (s.KeepRunning())
        std::cout << "The answer is " << 42 << ".\n";
}
BENCHMARK(ostream);

void print(benchmark::State& s) {
    while (s.KeepRunning())
        fmt::print("The answer is {}.\n", 42);
}
BENCHMARK(print);

void print_cout(benchmark::State& s) {
    std::ios::sync_with_stdio(false);
    while (s.KeepRunning())
        fmt::print(std::cout, "The answer is {}.\n", 42);
}
BENCHMARK(print_cout);

void print_cout_sync(benchmark::State& s) {
    std::ios::sync_with_stdio(true);
    while (s.KeepRunning())
        fmt::print(std::cout, "The answer is {}.\n", 42);
}
BENCHMARK(print_cout_sync);

BENCHMARK_MAIN();

```

The benchmark was compiled with Apple clang version 11.0.0 (clang-1100.0.33.17) with `-O3 -DNDEBUG` and run on macOS 10.15.4. Below are the results:

Run on (8 X 2800 MHz CPU s)

CPU Caches:

L1 Data 32K (x4)

L1 Instruction 32K (x4)

L2 Unified 262K (x4)

L3 Unified 8388K (x1)

Load Average: 1.83, 1.88, 1.82

Benchmark	Time	CPU	Iterations
printf	87.0 ns	86.9 ns	7834009
ostream	255 ns	255 ns	2746434
print	78.4 ns	78.3 ns	9095989
print_cout	89.4 ns	89.4 ns	7702973
print_cout_sync	91.5 ns	91.4 ns	7903889

Both `print` and `printf` are ~3 times faster than `cout` even with synchronization to the standard C streams turned off. `print` is 14% faster when printing to `stdout` than to `cout`. For this reason and because `print` doesn't use formatting facilities of `ostream` we propose using `stdout` as the default output stream and providing an overload for writing to `ostream`.

On Windows 10 with Visual C++ 2019 the results are similar although the difference between `print` writing to `stdout` and `cout` is smaller with `stdout` being 7% faster:

Run on (1 X 2808 MHz CPU)

CPU Caches:

L1 Data 32K (x1)

L1 Instruction 32K (x1)

L2 Unified 262K (x1)

L3 Unified 8388K (x1)

Benchmark	Time	CPU	Iterations
printf	835 ns	816 ns	746667
ostream	2410 ns	2400 ns	280000
print	580 ns	572 ns	1120000
print_cout	623 ns	614 ns	1120000
print_cout_sync	615 ns	614 ns	1120000

§ 7. Binary code

We propose minimizing per-call binary code size by applying the type erasure mechanism from [P0645]. In this approach all the formatting and printing logic is implemented in a non-variadic function `vprint`. Inline variadic

`print` function only constructs a `format_args` object, representing an array of type-erased argument references, and passes it to `vprint`:

```
void vprint(string_view fmt, format_args args);

template<class... Args>
inline void print(string_view fmt, const Args&... args) {
    return vprint(fmt, make_format_args(args...));
}
```

Below we compare the reference implementation of `print` to standard formatting facilities. All the code snippets are compiled with clang (Apple clang version 11.0.0 clang-1100.0.33.17) with `-O3 -DNDEBUG -c -std=c++17` and the resulting binaries are disassembled with `objdump -S`:

```
void printf_test(const char* name) {
    printf("Hello, %s!", name);
}
```

```
__Z11printf_testPKc:
    0:      55      pushq   %rbp
    1:      48 89 e5      movq    %rsp, %rbp
    4:      48 89 fe      movq    %rdi, %rsi
    7:      48 8d 3d 08 00 00 00      leaq    8(%rip), %rdi
    e:      31 c0      xorl    %eax, %eax
   10:      5d      popq    %rbp
   11:      e9 00 00 00 00      jmp     0 <__Z11printf_testPKc+0x16>
```

```
void ostream_test(const char* name) {
    std::cout << "Hello, " << name << "!";
}
```

__Z12ostream_testPKc:

```
0:      55      pushq   %rbp
1:      48 89 e5      movq   %rsp, %rbp
4:      41 56      pushq   %r14
6:      53      pushq   %rbx
7:      48 89 fb      movq   %rdi, %rbx
a:      48 8b 3d 00 00 00 00 movq   (%rip), %rdi
11:     48 8d 35 6c 03 00 00 leaq   876(%rip), %rsi
18:     ba 07 00 00 00 movl   $7, %edx
1d:     e8 00 00 00 00 callq  0 <__Z12ostream_testPKc+0x22>
22:     49 89 c6      movq   %rax, %r14
25:     48 89 df      movq   %rbx, %rdi
28:     e8 00 00 00 00 callq  0 <__Z12ostream_testPKc+0x2d>
2d:     4c 89 f7      movq   %r14, %rdi
30:     48 89 de      movq   %rbx, %rsi
33:     48 89 c2      movq   %rax, %rdx
36:     e8 00 00 00 00 callq  0 <__Z12ostream_testPKc+0x3b>
3b:     48 8d 35 4a 03 00 00 leaq   842(%rip), %rsi
42:     ba 01 00 00 00 movl   $1, %edx
47:     48 89 c7      movq   %rax, %rdi
4a:     5b      popq   %rbx
4b:     41 5e      popq   %r14
4d:     5d      popq   %rbp
4e:     e9 00 00 00 00 jmp    0 <__Z12ostream_testPKc+0x53>
53:     66 2e 0f 1f 84 00 00 00 00 00 nopw   %cs:(%rax,%rax)
5d:     0f 1f 00      nopl   (%rax)
```

```
void print_test(const char* name) {
    print("Hello, {}!", name);
}
```

__Z10print_testPKc:

```
0:      55      pushq   %rbp
1:      48 89 e5      movq   %rsp, %rbp
4:      48 83 ec 10     subq   $16, %rsp
8:      48 89 7d f0     movq   %rdi, -16(%rbp)
c:      48 8d 3d 19 00 00 00 leaq   25(%rip), %rdi
13:     48 8d 4d f0     leaq   -16(%rbp), %rcx
17:     be 0a 00 00 00 movl   $10, %esi
1c:     ba 0d 00 00 00 movl   $13, %edx
21:     e8 00 00 00 00 callq  0 <__Z10print_testPKc+0x26>
26:     48 83 c4 10     addq   $16, %rsp
2a:     5d      popq   %rbp
2b:     c3      retq
```

The code generated for the `print_test` function that uses the reference implementation of `print` described in this proposal is more than 2x smaller than the ostream code and has one function call instead of three. The `printf` code is further 2x smaller but doesn't have any error handling. Adding error handling would make its code size closer to that of `print`.

The following factors contribute to the difference in binary code size between `print` and `printf`:

- Passing format string as `string_view` instead of `const char*`.
- Capturing and passing argument type information.
- Preparing the array of formatting arguments.

§ 8. Impact on existing code

The current proposal adds new functions to the header `<format>` and should have no impact on existing code.

§ 9. Implementation

The proposed `print` function has been implemented in the the open-source fmt library [\[FMT\]](#) and has been in use for about 6 years.

§ 10. Wording

Add an entry for `__cpp_lib_print` to section "Header `<version>` synopsis [\[version.syn\]](#)", in a place that respects the table's current alphabetic order:

```
#define __cpp_lib_print 202005L **placeholder** // also in <format>
```

Modify section "Header `<format>` synopsis [\[format.syn\]](#)":

```
// 20.20.3, formatting functions
...
template<class... Args>
    size_t formatted_size(const locale& loc, wstring_view fmt, const Args&... args);

template<class... Args>
    void print(string_view fmt, const Args&... args);
template<class... Args>
    void print(FILE* stream, string_view fmt, const Args&... args);
template<class... Args>
    void print(ostream& os, string_view fmt, const Args&... args);

void vprint_unicode(string_view fmt, format_args args);
void vprint_unicode(FILE* stream, string_view fmt, format_args args);
void vprint_unicode(ostream& os, string_view fmt, format_args args);

void vprint_nonunicode(string_view fmt, format_args args);
void vprint_nonunicode(FILE* stream, string_view fmt, format_args args);
void vprint_nonunicode(ostream& os, string_view fmt, format_args args);

template<class... Args>
    void println(string_view fmt, const Args&... args);
```

Modify section "Formatting functions [\[format.functions\]](#)":

```
template<class... Args>
    size_t formatted_size(const locale& loc, wstring_view fmt, const Args&... args);
```

...

Throws: `format_error` if `fmt` is not a format string.

```
template<class... Args>
    void print(string_view fmt, const Args&... args);
```

Effects: Equivalent to:

```
print(stdout, fmt, make_format_args(args...));
```

```
template<class... Args>
    void print(FILE* stream, string_view fmt, const Args&... args);
```

Effects: If string literal encoding is UTF-8, equivalent to:

```
vprint_unicode(stream, fmt, make_format_args(args...));
```

Otherwise, equivalent to:

```
vprint_nonunicode(stream, fmt, make_format_args(args...));
```

```
template<class... Args>
void print(ostream& os, string_view fmt, const Args&... args);
```

Effects: If string literal encoding is UTF-8, equivalent to:

```
vprint_unicode(os, fmt, make_format_args(args...));
```

Otherwise, equivalent to:

```
vprint_nonunicode(os, fmt, make_format_args(args...));
```

```
void vprint_unicode(string_view fmt, format_args args);
```

Effects: Equivalent to:

```
vprint_unicode(stdout, fmt, args);
```

```
void vprint_unicode(FILE* stream, string_view fmt, format_args args);
```

Effects: Let `out = vformat(fmt, args)`. If `stream` refers to a terminal [Note: On POSIX and Windows meaning that `isatty(fileno(stream))` and `_isatty(_fileno(stream))` return 1 respectively. — end note] capable of displaying Unicode, writes `out` transcoded to the native system Unicode encoding to the terminal using the native API that preserves the encoding. Otherwise writes `out` to `stream` without transcoding.

Throws: `format_error` if `fmt` is not a format string, `std::system_error` if a call by the implementation to an operating system or other underlying API results in an error that prevents the function from meeting its specifications.

```
void vprint_unicode(ostream& os, string_view fmt, format_args args);
```

Effects: Let `out = vformat(os.getloc(), fmt, args)`. If `os` is a file stream (its associated stream buffer is an instance of `basic_filebuf`) that refers to a terminal capable of displaying Unicode, writes `out` transcoded to the native system Unicode encoding to the terminal using the native API that preserves the encoding. Otherwise

writes `out` to `stream` without transcoding.

Throws: `format_error` if `fmt` is not a format string, `std::system_error` if a call by the implementation to an operating system or other underlying API results in an error that prevents the function from meeting its specifications.

```
void vprint_nonunicode(string_view fmt, format_args args);
```

Effects: Equivalent to:

```
vprint_nonunicode(stdout, fmt, args);
```

```
void vprint_nonunicode(FILE* stream, string_view fmt, format_args args);
```

Effects: Writes the result of `vformat(fmt, args)` to `stream`.

Throws: `format_error` if `fmt` is not a format string, `std::system_error` if a call by the implementation to an operating system or other underlying API results in an error that prevents the function from meeting its specifications.

```
void vprint_nonunicode(ostream& os, string_view fmt, format_args args);
```

Effects: Writes the result of `vformat(os.getloc(), fmt, args)` to `os`.

Throws: `format_error` if `fmt` is not a format string, `std::system_error` if a call by the implementation to an operating system or other underlying API results in an error that prevents the function from meeting its specifications.

```
template<class... Args>
void println(string_view fmt, const Args&... args);
```

Effects: Equivalent to:

```
print("{}\n", format(fmt, args...));
```

§ 11. Acknowledgements

Thanks to Corentin Jabot for his work on text encodings in C++ and in particular [\[P1885\]](#) that will simplify implementation of the current proposal.

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